B.Tech CSE - Third Year | Lab 2

Assignment 2: Largest, Smallest and Count • *Aim: Find extrema and count values that satisfy a condition.*

Experiment No. 2.1: Find largest array value

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_01_Find_largest_array_value {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int largest = marks[0];
        for (int i = 1; i < marks.length; i++) {
            if (marks[i] > largest) largest = marks[i];
        }
        System.out.println("Largest: " + largest);
    }
}
```

3. **Output:** `Largest: 90`
4. **Viva-Voce:** Why start with marks[0]? What happens if the array is empty? What is the time complexity?

Experiment No. 2.2: Find smallest array value

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_02_Find_smallest_array_value {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int smallest = marks[0];
        for (int i = 1; i < marks.length; i++) {
            if (marks[i] < smallest) smallest = marks[i];
        }
        System.out.println("Smallest: " + smallest);
    }
}
```

3. **Output:** `Smallest: 45`
4. **Viva-Voce:** Which comparison finds a smaller value? Why is initialization important? Can this work with negative values?

Experiment No. 2.3: Count values greater than 60

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_03_Count_values_greater_than_60 {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int count = 0;
        for (int value : marks) {
            if (value > 60) count++;
        }
        System.out.println("Count: " + count);
    }
}
```

3. **Output:** `Count: 3`
4. **Viva-Voce:** What is a counter? Does 60 itself satisfy value > 60? What is the difference between count and sum?